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Design Challenges and Techniques for 5nm FinFET CMOS Analog/Mixed-Signal Circuits

Saurabh Goyal[1], Sanjay Kumar Wadhwa[2], Divya Tripathi[3]
Gaurav Agrawal[4], Krishna Thakur[5], Deependra Kumar Jain[6]
Alvin L.S. Loke[7], Atul Kumar[8], Manish Kumar Upadhyay[9]
NXP Semiconductors, India
[1]saurabh.goyal@nxp.com, [2]sanjay.wadhwa@nxp.com, [3]divya.tripathi@nxp.com
[4]gaurav.agrawal@nxp.com, [5]krishna.thakur@nxp.com, [6]deependra.jain@nxp.com
[7]alvin.loke@nxp.com, [8]atul.kumar_1@nxp.com,[9]manish.upadhyay@nxp.com
Bhawna[10], [10]1607_bhawna@gmail.com, Western Digital, India
Sanjoy Kumar Dey[11], Intel Corporation, India, [11]sanjoy.kumar.dey@intel.com

***Abstract*— Recent CMOS scaling incessantly focuses on improving density, speed, area, and power consumption in digital circuits which is increasingly less applicable to analog/mixed-signal circuits. This paper proposes various practical techniques to overcome technology challenges in several common analog/mixed-signal circuits in 5nm CMOS. Issues addressed include high leakage in dummy devices due to continuous active area layout style, large parasitics in ring oscillators, and gate-induced drain leakage in SAR ADC sampling switches.**

Keywords—5nm finFET technology, leakage currents, gate-induced drain leakage, ring oscillators

I. Introduction

CMOS technology scaling has provided many advantages to the latest IC designs. Despite asymptotic physical feature scaling, design/technology co-optimization in the most advanced nodes continues to offer improved device area density, speed, area, and power to enable unprecedented levels of system performance and integration. Scaling benefits digital circuits but plague analog/mixed-signal circuits with many challenges. For instance, MOS threshold voltage (V_T) is not scaling as aggressively as supply voltage. The scaling of the transistor gate length contributes to leakage. Furthermore, layout rules have become ever more restrictive to ensure manufacturing yield and contain device variation. A plethora of layout techniques have addressed some of these challenges [1]–[4]. For example, [1] proposes a method to reduce leakage currents using intra-gate biasing. The width and length of the transistor gate have been adjusted to increase VT and reduce leakage, but this scheme is not applicable as jogged active areas are discouraged in 5nm. Inverter layout techniques in [2] reduce inverter propagation delays by decreasing the routing length of the input and output nodes. Ref [4] presents a multi-finger layout technique for better matching and performance. This technique reduces the transistor's intrinsic resistance but does little to reduce parasitic resistance and capacitance.

This paper addresses several common issues faced when porting analog/mixed-signal designs from older nodes to 5nm CMOS: leakage in dummy transistors, interconnect parasitics in ring oscillators, and gate-induced drain leakage in SAR ADC sampling switches. We introduce layout techniques to reduce leakage as well as parasitic resistance and capacitance, especially significant in the middle-end-of-line and lower back-end-of-line. Section II describes the impact of leakage current flowing in mandatory dummy transistors around the active devices. Section III addresses the impact of parasitics in high-performance ring oscillators. Section IV describes the impact of high gate-induced drain leakage in SAR ADC sampling switches.

II. Overcoming Leakage to Dummy Transistors

Continuous active area layout styles are increasingly common for device density and performance improvement. This layout style establishes a stress plateau where the mobility of devices is stable and maximal [5]. Electrically deactivated dummy devices are inserted in between two active devices to maintain continuity in the active area. The NMOS and PMOS dummy device terminals are connected to ground and supply, respectively. These dummy devices around the active devices contribute to leakage, especially at the fast process corner and at high temperatures, thereby degrading the design performance.

Two possibilities exist for connecting dummy transistors around the active devices. The first is to connect the dummy devices (MD) in the centre as shown in Fig. 1. The gates of the MOS and PMOS dummies are connected to VSS and VDD, respectively, making them electrically deactivated. When Sig1 and Sig2 voltages are different, subthreshold leakage can flow. Another possibility is connecting the dummy transistors at the end. The Gate/Source/Drain of NMOS and PMOS dummies are connected to VSS and VDD, respectively. Leakage flows from Sig1/Sig2 to VDD/VSS through the dummies as shown in Fig. 2.

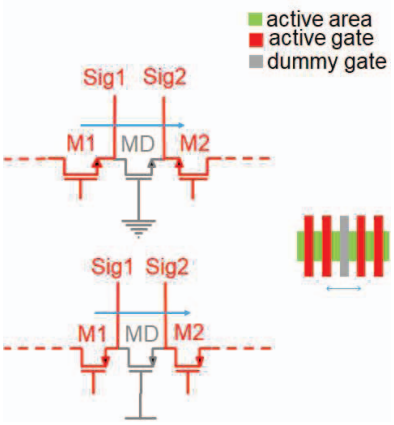


Fig.1. Dummy transistors in the center of a continuous active area.

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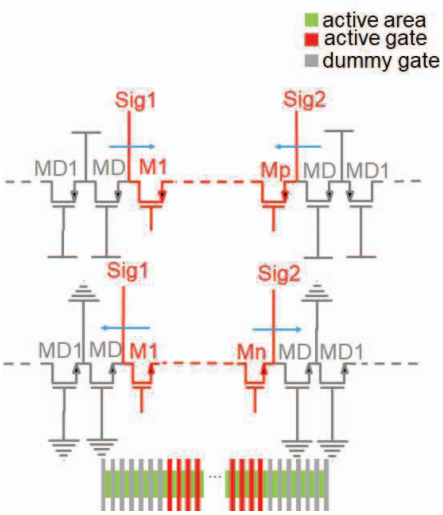


Fig.2. Dummy transistors at the end of an active area.

Fig. 3 illustrates two methods to reduce the leakage currents to the dummy transistors. In the first method, the diffusion of dummy transistors is connected to signal nets Sig1/Sig2 as shown by the green dashed line. This method adds more capacitance to the signal nets, which reduces the speed of operation. This results in a lower frequency of changing the trim bits in the resistor ladder. The second method is to insert a larger number of dummy devices. By floating the intermediate diffusion nodes of the series dummies, the effective channel length of the dummy device tied to Sig1/Sig2 is significantly increased, thereby dramatically reducing leakage. The length of dummy devices will be N times the length of a single transistor connected as a dummy device, where N is the total number of transistors.

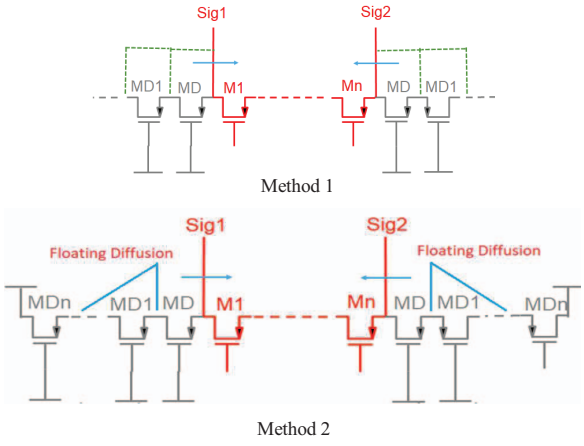


Fig. 3. Methods to reduce leakage drawn by adjacent dummy devices.

To illustrate the significance of leakage to dummies and our proposed methods to reduce this leakage, consider a resistor ladder as shown in Fig 4 for generating a reference voltage (V_{ref}) that dictates the frequency of a free-running oscillator (FRO). The current flowing through the resistor ladder is 2 µA. Six bits exist for fine trimming, so 64 switches are tapped along the ladder to select the desired V_{ref}.

$$V_{ref} = \frac{VDD_CORE*150K}{(COARSEBIT*10.7K+FINEBIT*538+75K)}$$

Dummy devices surround the active devices to protect the active devices from variations due to density gradients [6]. The leakage current ($I_{leakage}$) at fast processes and hot corners drawn by these dummies becomes a significant portion of the total current consumed by the circuit. As shown in Table I, the leakage current in simulations becomes 14% of the active current flowing in the resistor ladder, subsequently disturbing V_{ref} at the hot temperature as shown in Table II. This results in an unacceptably large 10% frequency deviation that violates the design specification. With the improved layout as per the Method II shown in Fig 3, the leakage current reduces to only 2.9% of the active ladder current, causing less than 1-mV of V_{ref} deviation across temperature.

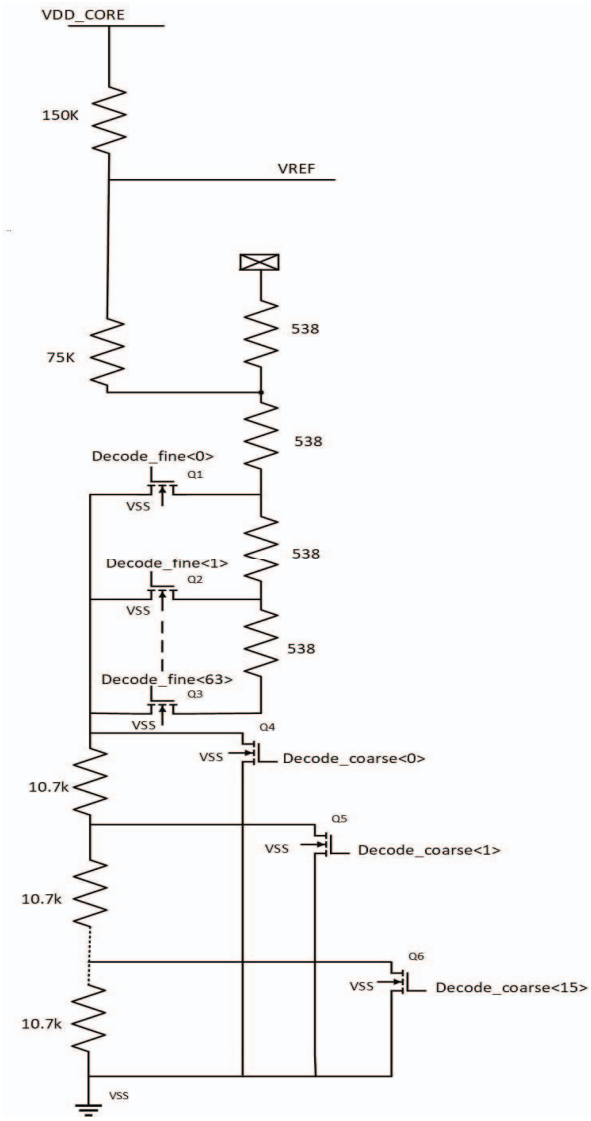


Fig.4. Example of a Resistance Ladder to generate the reference voltage

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TABLE I. Worst-case Leakage Current

Supply Voltage	MOS Corner	Resistor Corner	Layout	$I_{leakage}$		
				–40 °C	27 °C	125 °C
0.825 V	FF	FF	Initial	0.3 nA	5.2 nA	280 nA
			Improved	0.2 nA	2.6 nA	57 nA

TABLE II. Worst-case Reference Voltage

Supply Voltage	MOS Corner	Resistor Corner	Layout	V_{ref}		
				–40 °C	27 °C	125 °C
0.825 V	FF	FF	Initial	0.540 V	0.540 V	0.527 V
			Improved	0.541V	0.541 V	0.540 V

III. Overcoming Parasitics in Ring Oscillator

Given the compact, three-dimensional nature of the finFET structure and surrounding interconnect, parasitics—especially resistance—are increasing at a rapid rate. See Fig. 5. High parasitic resistance leads to supply drop, which significantly degrades FET performance, especially at the low supply voltage commonplace in finFET nodes. In addition, high parasitic capacitance due to strong gate coupling to diffusion contacts and overlying metal leads to many design iterations to reconcile the gap between initial schematic-based (pre-layout) and layout-based (post-layout) targets [7]. Consider the five-stage ring oscillator in Fig. 6, designed for 7.0 GHz. The oscillator is current-starved with a maximum amplitude (V_{ring}) of 0.825 V to avoid electrical overstress. A current of 2 mA was forced from an ideal source. Table III shows the gap between pre- and initial post-layout simulations. In this comparison, some estimated parasitics were already included in the schematic-based simulation. So, to achieve the 7.0 GHz frequency in post-layout, higher current has to be forced, resulting in V_{ring} exceeding the reliability limit.

To achieve the 7.0-GHz target, five major modifications were made to the layout. First, the two inverters highlighted in Fig.6 are placed together with a shared active area. With this continuous active area layout style layout, device mobility is maximized and constant for optimum matching and performance. The length of oxide definition (LOD) effect saturates after a sufficient number of dummy devices are appended to the edges of the continuous active area [7]. Secondly, minimum-length devices with slightly larger source/drain areas are used for reduced contact resistance and improved drive current.

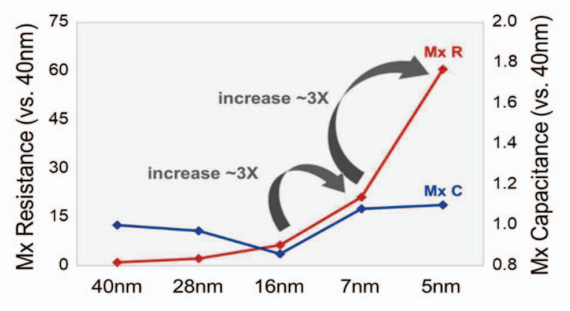


Fig. 5. Interconnect parasitics with technology scaling [5].

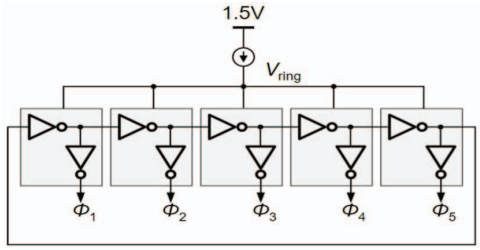


Fig. 6. Five-stage inverter-based ring oscillator.

TABLE III. Pre- and Post-layout Simulation Results

Design Criterion	*Pre-layout Simulation*	*Post-layout Simulation*
Current	2 mA	2 mA
Frequency	7.13 GHz	5.85 GHz
V_{ring}	0.666 V	0.804 V

Third, the usage of the Metal-0 (M0) metal layer over the MOSFET gates is minimized to reduce severe capacitance. M0 was mainly used for layer promotion to higher metal layers that exhibit far fewer parasitics. This is done by reducing the number of M0 routing tracks to 3-track M0, exchanging routing density for reduced parasitics. A wider M0 is used to further reduce resistance. A comparison is shown in Fig. 7. With the 3-track M0, V_{ring} is now reduced to 0.775 V and the frequency increased to 6.37 GHz, an almost 9% improvement. Fourth, the overhang of the diffusion contacts beyond the active area is minimized to reduce contact-to-gate capacitance [11]. The aspect ratio and floorplan of the oscillator's five stages are reworked to shorten inter-stage routes, as shown in Fig. 8. This helps in reducing the parasitic RC on the five phases as well as helps in matching their route length. A higher and wider metal layer is used for inter-stage routing, which helps in better parasitic resistance. The performance of several layout iterations is summarized in Fig. 9. With all of the above layout improvements, the ring oscillator can now run at 7.0 GHz and meet design specifications. See Table IV.

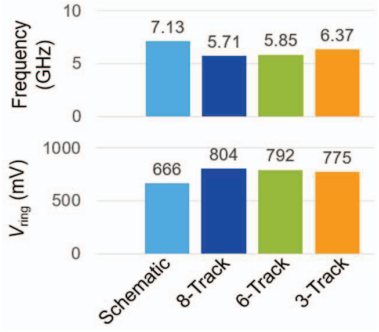


Fig. 7. Oscillator performance vs. number of M0 tracks.

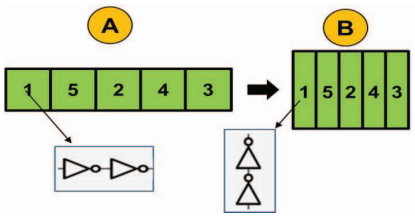


Fig. 8. Aspect ratio optimization to reduce parasitic.

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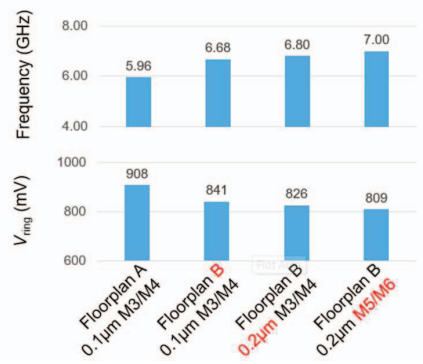


Fig.9. Comparison with floorplan A, B and with the use of higher and wider metal layer.

TABLE IV. Alignment between Pre- and Post-layout Simulations

Simulation Comparison			
Design Criterion	***Schematic Simulation***	***Initial Layout Simulation***	***Optimized Layout Simulation***
Current	2mA	2mA	**2mA**
Frequency	7.13GHz	5.85GHz	**7.01GHz**
Vring	0.666V	0.804V	**0.809V**

IV. Overcoming Gate-Induced Drain/Source Leakage

When a MOSFET is biased in the off state and voltage at the drain/source is high, significant drain to body/source to body leakage current can be observed (Fig. 11). GIDL/GISL currents result from tunneling of valence band electrons to the conduction band at the overlap region between gate and drain/source where strong vertical and longitudinal fields are present(Fig. 10). Mitigating this effect in 5nm CMOS is difficult. GIDL/GISL current increases on lowering the gate voltage because of the higher voltage difference between the gate and drain/source. This makes conventional leakage reduction techniques, e.g., negative charge pumps, much less effective to reduce GIDL.

The setup shown in Fig. 12 simulates 5nm device leakage. NMOS and PMOS are biased in the off state, a common-mode voltage (V_{cm}) is connected to the drain, and a voltage (Vx) connected to the source is swept from zero to supply.

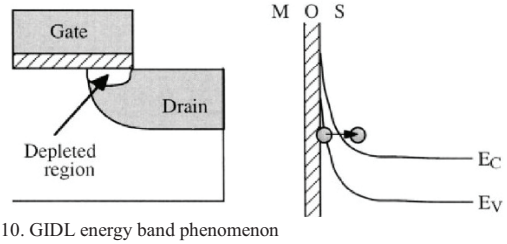


Fig. 10. GIDL energy band phenomenon

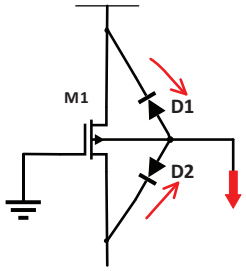


Fig. 11. Gate-induced drain leakage phenomenon.

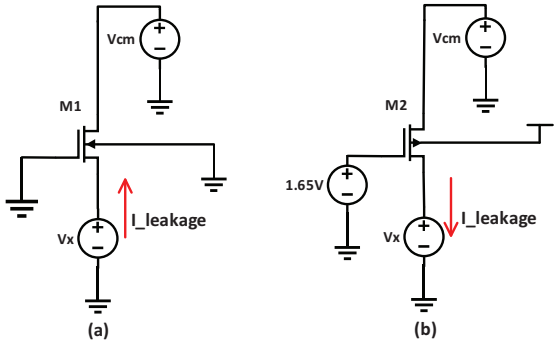


Fig. 12. Setup for (a) leakage in NMOS and (b) leakage in PMOS.

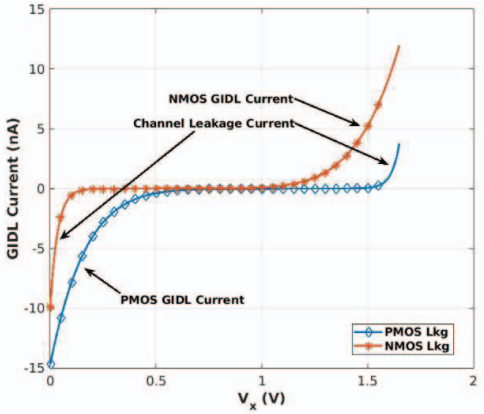


Fig. 13. NMOS and PMOS leakage in 5nm.

In the NMOS and PMOS leakage characteristics shown in Fig. 13, GIDL current increases exponentially for NMOS above 1.2 V and dominates below 0.4 V for PMOS. Controlling the device off-state leakage is critical for the performance of many mixed-signal blocks, including SAR ADC.

There are two phases in the SAR ADC operation: sampling and conversion. Fig. 14 shows the capacitor configuration in the SAR ADC. During the sampling phase, the input voltage (V_{in}) is sampled onto a sampling capacitor (C_s) by closing the input and top plate switches. During the conversion phase, the input switch and top plate switches are open. The charge stored in the sampling phase is redistributed among the capacitors in the DAC by opening and closing switches sw1/sw2.

The top plate of the capacitors is undriven in the conversion phase and any charge leakage through the top plate switch

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will change the value of the charge stored on the capacitor, resulting in an error. Due to the nonlinear nature of GIDL current, the error will be nonlinear and contribute to the DAC INL. The issue will be more pronounced for Synchronous SAR ADCs working on a slow clock. If the top plate switch is implemented with NMOS, the error will be highest when the top plate voltage is full-scale and negligible below 75 % of the full-scale input. If the switch is implemented with PMOS, the error will be highest when the top-plate voltage is zero and negligible above 400 mV. If the switch is implemented using a CMOS transmission gate, then the error would be maximum near zero and full-scale inputs and negligible near the mid-scale.

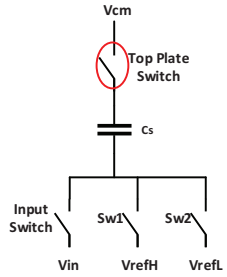


Fig. 14. Typical capacitor configuration in SAR ADC.

In 5nm, GIDL becomes significant for NMOS (biased in off state) at a drain voltage greater than 1.2 V and increases exponentially with increasing drain voltage. We can reduce the leakage by disallowing the voltage between the gate and drain/source to exceed 1.2 V. In Fig. 15, we have proposed a scheme to eliminate GIDL current during the conversion phase of a SAR ADC. By sensing whenever node A exceeds 1.2V, we can increase the gate potential by around 400 mV to lower the gate-to-drain voltage and thereby reduce the GIDL current. The Sense block is an amplifier followed by a Schmitt trigger for some built-in hysteresis. The sense block (S) output feeds a digital control block that selects the gate bias for the switch (M1) depending on the ADC mode and potential at node A. When the ADC is in the sampling phase, the switch is turned ON to apply common-mode voltage to the capacitor top plate. In the conversion phase, if the potential at node A is less than 1.2 V, S will be zero and the switch will be turned off with zero volts applied to the gate terminal. When the potential at node A exceeds 1.2 V, then the switch is turned off by applying around 400 mV at its gate. This will not allow the gate-to-drain voltage to go beyond 1.2 V and thereby limit the impact of GIDL. A similar scheme can be used for the PMOS switch. In Fig. 16, we have shown that when the proposed GIDL solution is enabled (for an NMOS switch), leakage is dramatically reduced when the top-plate voltage exceeds 1.2 V. Fig. 17 demonstrates the impact of the proposed GIDL solution on the error in sampled voltage. When the GIDL solution is enabled, there is no change in the top plate voltage during conversion.

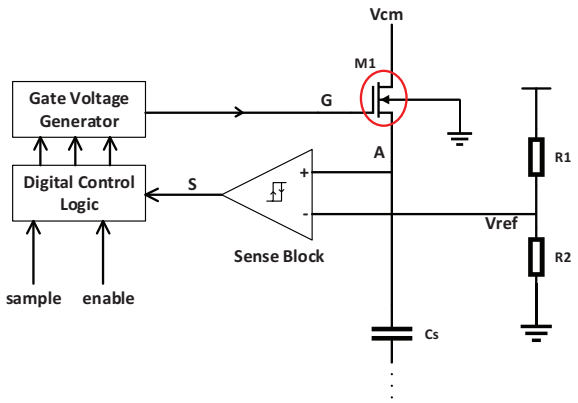


Fig. 15. Proposed solution to reduce GIDL in SAR ADC.

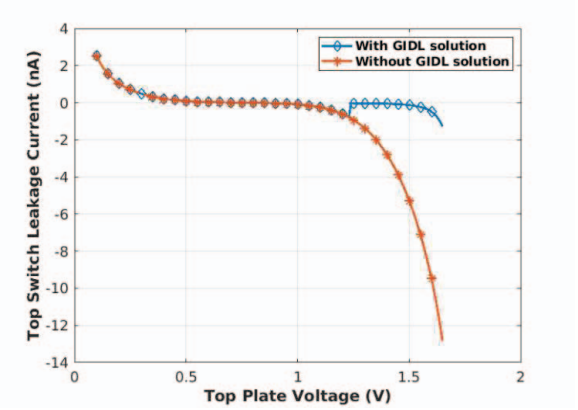


Fig. 16. NMOS Leakage with and without GIDL solution.

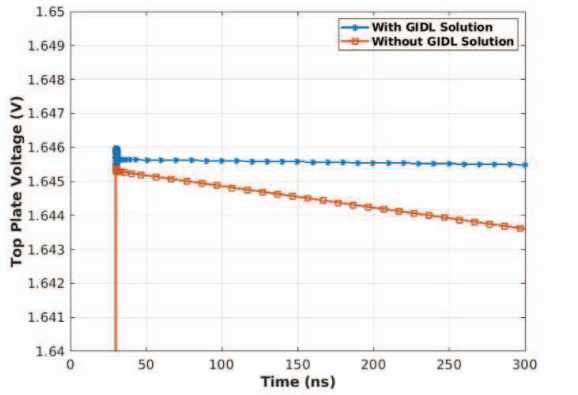


Fig. 17. Top Plate Voltage droop in conversion phase with and without GIDL Solution

V. Conclusions

This paper presented some key challenges in analog/mixed-signal designs as we scale to 5nm CMOS technology. This paper discussed how to reduce leakage currents in dummy transistors, interconnect parasitics to maximize ring oscillator performance and GIDL. These techniques were effective in overcoming issues imposed by the technology in order to meet the required design performance.

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